Service Bulletin

Service Bulletin Number: 3810451	Released Date: 01-Nov-2011
Bearing Corrosion	

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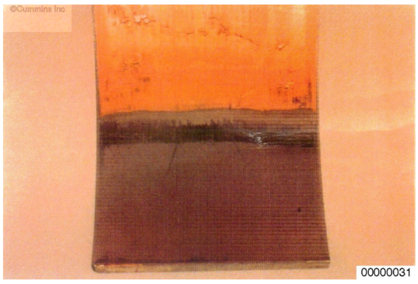
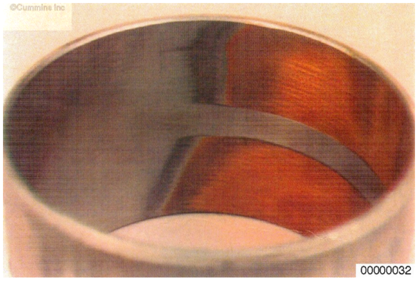
Coolant in the lubricating oil can cause bearing corrosion. Organic acids from coolant breakdown form in the lubricating oil and, given enough time, will begin to attack and leach the lead out of copper/lead bearing materials. As the lead is leached from the bearing the remaining copper begins to fatigue and progresses to failure. Reference the illustration.



Beginning in the early 1980's, oil formulations were changed to make oil more moisture tolerant. Coolant is better dispersed in the lubricating oil, making it more difficult to detect.

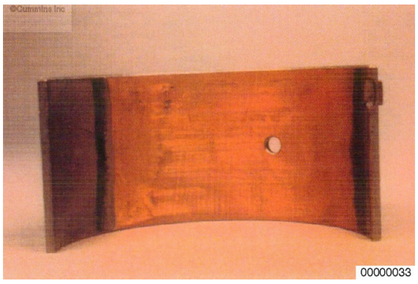
Lubricating oil analysis is an effective way of detecting the presence of coolant in the oil. Oil analysis results that indicate 50 ppm, (or greater), sodium or potassium, will indicate the presence of coolant. If coolant is present and the same oil analysis results indicate 50 ppm, (or greater) copper or lead the the new oil, an engine bearing inspection needs to be scheduled.

A bearing that has been subjected to acidic attack long enough to sustain damage will have a “frosty” appearance in the exposed copper region. Any bearing exhibiting corrosive damage **must** be replaced. Reference the illustration.



Acidic attack of bearings is accelerated by high load/stress, so those bearings which have the highest unit loading/stress will show damage first. Engine bearings will show corrosive damage in the following order. Reference the illustration.

1. Rod bearings
2. Piston pin bushings
3. Camshaft bushings
4. Main bearings



5. Accessory drive thrust bearings
6. Camshaft gear thrust bearings
7. Accessory drive bearings
8. Camshaft follower pins.

Document History

Date	Details
2011-10-31	Module Created
2011-10-31	To make existing Service Bulletin 3810451 available on QSOL with color pictures.
2011-11-7	QSOL Quick Fix|Reason: Outfile|Notes: none
2012-5-27	Ticket No.: 39439|Reason: QSOL Display Problem|Notes: SB is not appearing for any engine search on QSOL.

Last Modified: 27-May-2012